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Static and Dynamic (RIP) Routing between Two Routers in GNS3

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Aim

To configure static routes on two routers in GNS3 so that two different networks (192.168.1.0/24 and 192.168.2.0/24) can communicate with each other, and then, on the same network, to replace the static routes with dynamic routing using RIP and verify the connectivity.

Requirements

GNS3 with Cisco router images (2 routers having FastEthernet ports), 2 Ethernet switches and 2 VPCS (Virtual PC Simulator) nodes.

IP Addressing Table

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC1 – e0	192.168.1.2	255.255.255.0	192.168.1.1
PC2 – e0	192.168.2.2	255.255.255.0	192.168.2.1
R1 – Fa0/0	10.1.1.1	255.0.0.0	—
R1 – Fa0/1	192.168.1.1	255.255.255.0	—
R2 – Fa0/0	10.1.1.2	255.0.0.0	—
R2 – Fa0/1	192.168.2.1	255.255.255.0	—

PART A: STATIC ROUTING

1. Build the Topology and Start the Devices

Step 1: Open GNS3 and create a new project.

Step 2: Drag two routers (R1, R2), two Ethernet switches (Switch1, Switch2) and two VPCS (PC1, PC2) to the workspace.

Step 3: Connect the devices using the Add a link tool: R1 Fa0/0 – R2 Fa0/0; R1 Fa0/1 – Switch1 e0; Switch1 e1 – PC1 e0; R2 Fa0/1 – Switch2 e0; Switch2 e1 – PC2 e0.

Step 4: Click the green Start button to start all the devices. Wait until all the devices are running.

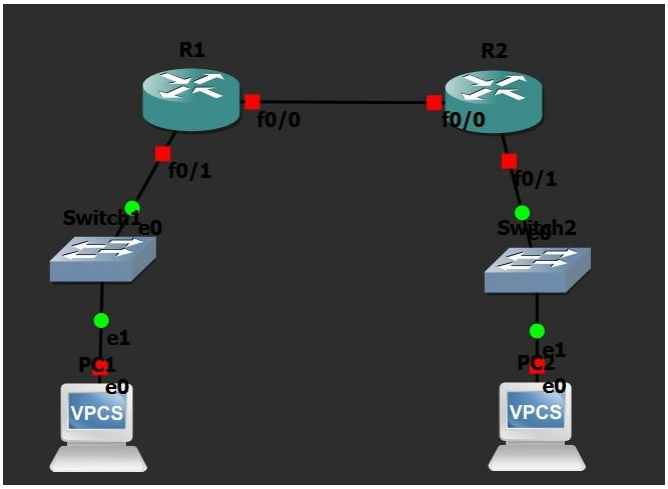


Fig 1: Topology of the experiment in GNS3

2. Configure the PCs

Right-click the PC → Console. In the VPCS console, use the ip command in the format: ip <address> <mask> <gateway>.

Step 5: PC1: set IP 192.168.1.2, mask 255.255.255.0 and gateway 192.168.1.1.

```
PC1> ip 192.168.1.2 255.255.255.0 192.168.1.1
```

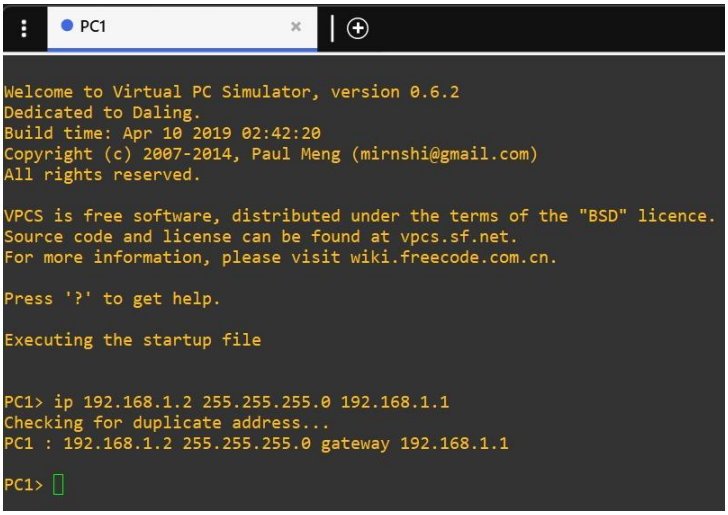


Fig 2: IP configuration of PC1

Step 6: PC2: set IP 192.168.2.2, mask 255.255.255.0 and gateway 192.168.2.1.

```
PC2> ip 192.168.2.2 255.255.255.0 192.168.2.1
```

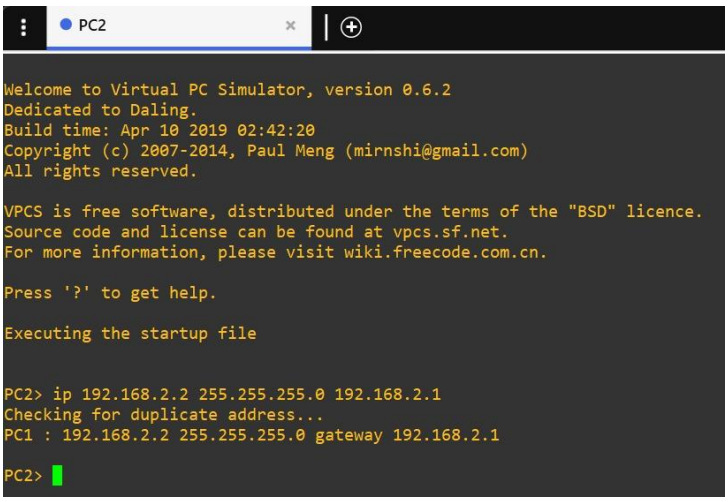


Fig 3: IP configuration of PC2

3. Configure the Router Interfaces

Right-click the router → Console. Use the commands below; no shutdown turns the interface on.

Step 7: R1: Fa0/0 = 10.1.1.1 / 255.0.0.0 and Fa0/1 = 192.168.1.1 / 255.255.255.0.

```
R1# conf t
R1(config)# interface fa 0/0
R1(config-if)# ip address 10.1.1.1 255.0.0.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# conf t
R1(config)# interface fa 0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# show ip interface brief
```

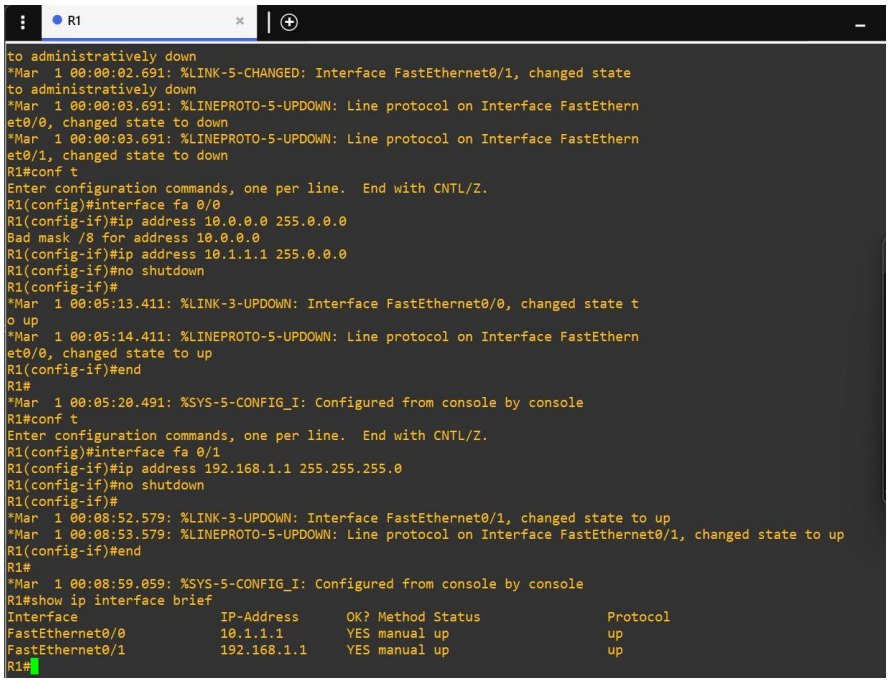


Fig 4: R1 interface configuration and show ip interface brief

Note: 10.0.0.0 with mask 255.0.0.0 gave “Bad mask /8 for address 10.0.0.0”, because 10.0.0.0 is the network address and cannot be given to an interface. A host address (10.1.1.1) is used instead.

Step 8: R2: Fa0/0 = 10.1.1.2 / 255.0.0.0 and Fa0/1 = 192.168.2.1 / 255.255.255.0.

```
R2# conf t
R2(config)# interface fa 0/0
R2(config-if)# ip address 10.1.1.2 255.0.0.0
R2(config-if)# no shutdown
R2(config-if)# end
R2# conf t
R2(config)# interface fa 0/1
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# end
```

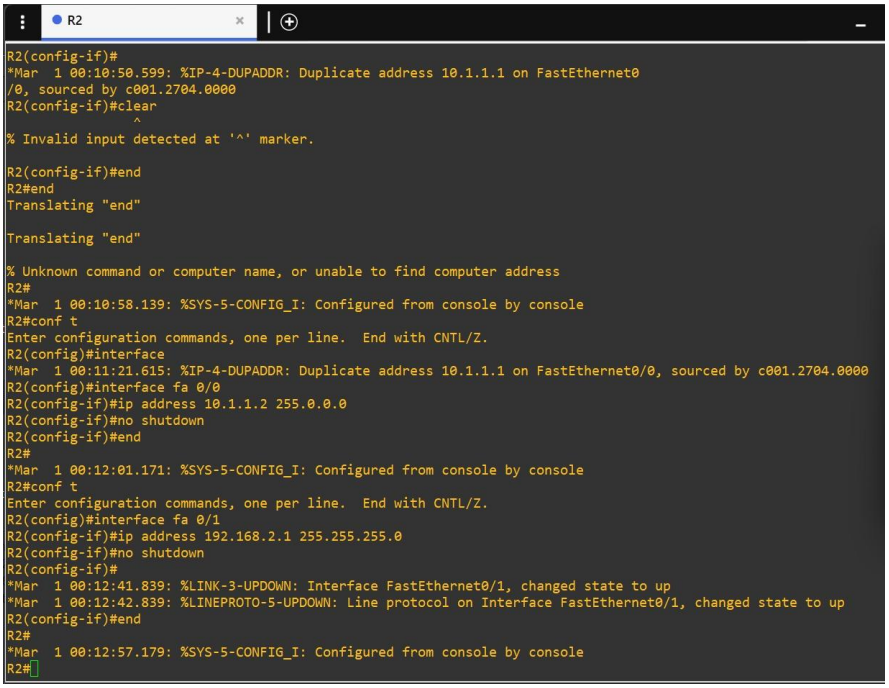


Fig 5: R2 interface configuration

Note: The message "Duplicate address 10.1.1.1 on FastEthernet0/0" appeared because the address 10.1.1.1, which already belongs to R1, was also in use on the link from R2's side. It was solved by giving R2 the address 10.1.1.2 (both ends of a link need different addresses in the same network).

Step 9: Verify the interfaces on R2 using show ip interface brief. Both interfaces must be up/up.

```
R2# show ip interface brief
```

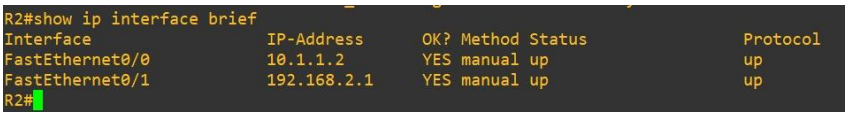


Fig 6: show ip interface brief on R2

4. Configure Static Routes

Each router only knows its directly connected networks. A static route tells it how to reach the other router's LAN. Format: ip route <destination network> <mask> <next-hop IP>.

Step 10: On R1, add a route to 192.168.2.0 through the next hop 10.1.1.2 (R2).

```
R1# conf t
R1(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.2
R1(config)# end
```

Step 11: Check the routing table on R1. The new route is shown with the code S (static); the connected networks are shown with C.

```
R1# show ip route
```

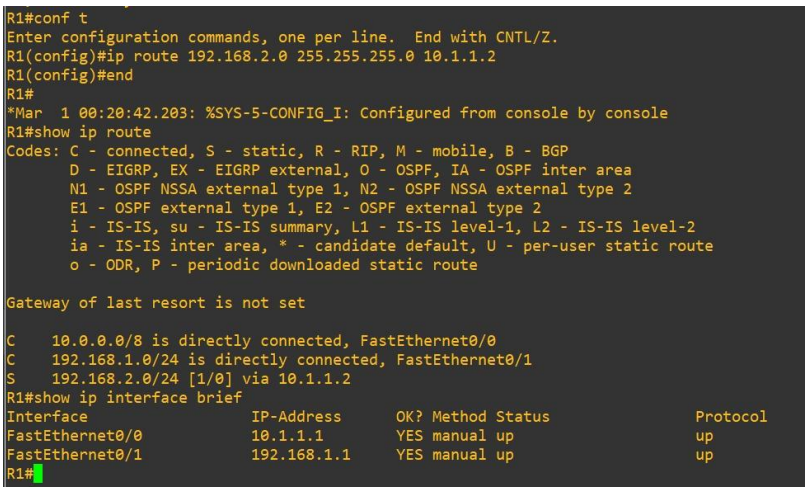


Fig 7: show ip route on R1 – static route 192.168.2.0/24 via 10.1.1.2

Step 12: On R2, add the return route to 192.168.1.0 through the next hop 10.1.1.1 (R1). Without this, the replies cannot return.

```
R2# conf t
R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)# end
R2# show ip route
```

Verification of Static Routing

Step 13: On R2, show ip route must show S 192.168.1.0/24 [1/0] via 10.1.1.1, together with the connected networks C 10.0.0.0/8 and C 192.168.2.0/24.

Step 14: From PC1 console ping PC2 (and from PC2 ping PC1). Replies mean the static routing works. The first packet may time out because of ARP.

```
PC1> ping 192.168.2.2
PC2> ping 192.168.1.2
```

Step 15: Optionally run trace from PC1 to see the path through R1 (192.168.1.1) and R2 (10.1.1.2).

```
PC1> trace 192.168.2.2
```

Note: the ping and R2 route steps above have no screenshots in the photo set; add your own screenshots at this place.

Part A result: Static routes on R1 and R2 allow PC1 and PC2 to communicate.

Quick Start Summary – Build Everything from Scratch

Use this block only if you want to start Dynamic Routing on a completely new network. If you are continuing from Part A (same network), skip this block and start Part B at "B1. Remove the Static Routes". If you start from scratch, skip B1 and go directly to "B2. Configure RIP".

Stage	What to do
1. Topology	Place R1, R2, Switch1, Switch2, PC1, PC2. Links: R1 Fa0/0–R2 Fa0/0; R1 Fa0/1–Switch1 e0; Switch1 e1–PC1 e0; R2 Fa0/1–Switch2 e0; Switch2 e1–PC2 e0. Start all devices.
2. PC IPs	Set the IP, mask and gateway on PC1 and PC2 (commands below).
3. Router IPs	Give the interfaces of R1 and R2 their IPs and turn them on with no shutdown (commands below).
4. Check	show ip interface brief on both routers – all interfaces up/up.
5. Dynamic routing	Continue with Part B from "B2. Configure RIP". No static route is needed.

```
PC1> ip 192.168.1.2 255.255.255.0 192.168.1.1
PC2> ip 192.168.2.2 255.255.255.0 192.168.2.1

R1# conf t
R1(config)# interface fa 0/0
R1(config-if)# ip address 10.1.1.1 255.0.0.0
R1(config-if)# no shutdown
R1(config-if)# interface fa 0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end

R2# conf t
R2(config)# interface fa 0/0
R2(config-if)# ip address 10.1.1.2 255.0.0.0
R2(config-if)# no shutdown
R2(config-if)# interface fa 0/1
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# end
```

PART B: DYNAMIC ROUTING (RIP)

The same network and IP addresses of Part A are used. Instead of typing a route on every router, RIP (Routing Information Protocol) is enabled so that R1 and R2 learn each other's LANs automatically.

B1. Remove the Static Routes

The static routes are deleted first, so that the routes are learned only through RIP (a static route has a lower administrative distance and would hide the RIP route). Skip this part if you started from scratch.

Step 16: On R1, remove the static route and check the routing table. show commands are given in privileged mode (R1#), so type end first; typing show ip route in configuration mode gives "Invalid input". Only the connected (C) networks must remain.

```
R1# conf t
R1(config)# no ip route 192.168.2.0 255.255.255.0 10.1.1.2
R1(config)# end
R1# show ip route
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 192.168.2.0 255.255.255.0 10.1.1.2
R1(config)#show ip route
           ^
% Invalid input detected at '^' marker.

R1(config)#end
R1#
*Mar  1 00:50:13.979: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
R1#
```

Fig 8: Static route removed on R1 – only connected routes remain

Step 17: On R2, remove its static route and check the routing table in the same way.

```
R2# conf t
R2(config)# no ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)# end
R2# show ip route
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)#end
R2#show
*Mar  1 00:50:50.327: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
R2#
```

Fig 9: Static route removed on R2 – only connected routes remain

B2. Configure RIP

Step 18: On R1, enable RIP and advertise the directly connected networks 10.1.1.0 and 192.168.1.0. (no router rip first clears any old RIP configuration; it is optional.)

```
R1# conf t
R1(config)# no router rip
R1(config)# router rip
R1(config-router)# network 10.1.1.0
R1(config-router)# network 192.168.1.0
R1(config-router)# end
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no router rip
R1(config)#end
R1#
*Mar  1 01:12:26.019: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#network 10.1.1.0
R1(config-router)#network 192.168.1.0
R1(config-router)#end
R1#
*Mar  1 01:13:10.287: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Fig 10: RIP configuration on R1

Step 19: On R2, enable RIP and advertise 10.1.1.0 and 192.168.2.0.

```
R2# conf t
R2(config)# router rip
R2(config-router)# network 10.1.1.0
R2(config-router)# network 192.168.2.0
R2(config-router)# end
```

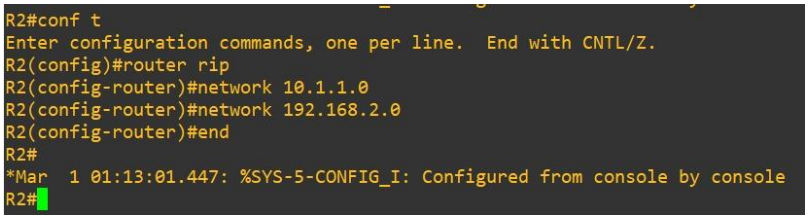


Fig 11: RIP configuration on R2

Note: No version command was given, so RIP version 1 is used (the debug output shows "v1 update"). RIPv1 is classful, so the network 10.1.1.0 is taken as 10.0.0.0.

B3. Observe the RIP Updates

Step 20: On R1, turn on RIP debugging. R1 sends RIP updates every 30 seconds; the output shows the networks and their metric (hop count) being advertised.

```
R1# debug ip rip
```

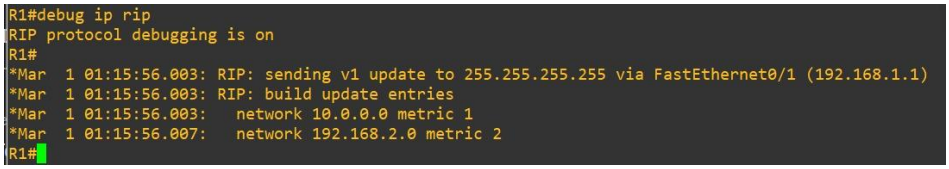


Fig 12: debug ip rip on R1 – update sent through Fa0/1

Step 21: On R2, turn on RIP debugging. R2 receives the update from R1 (10.1.1.1) that tells it 192.168.1.0 is 1 hop away, and sends its own updates.

```
R2# debug ip rip
```

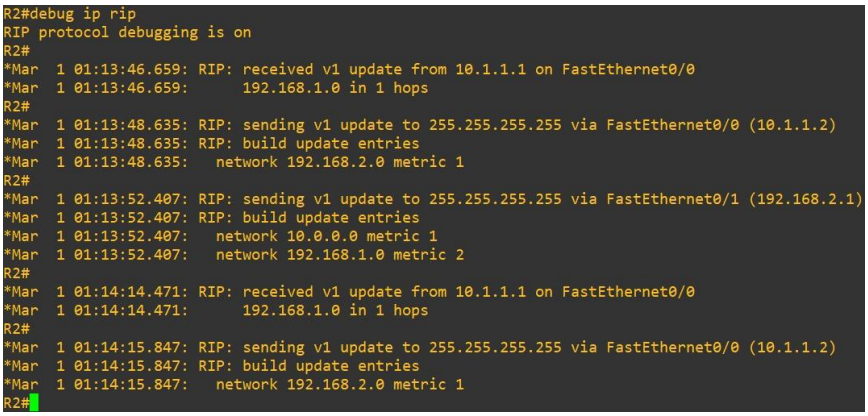


Fig 13: debug ip rip on R2 – update received from R1

B4. Verify Connectivity

Step 22: From PC1 ping PC2. The first packets may time out while the routers exchange their first updates and ARP is resolved; after that the replies arrive (ttl=62, because the packets crossed two routers).

```
PC1> ping 192.168.2.2
```

```
PC1> ping 192.168.2.2
192.168.2.2 icmp_seq=1 timeout
192.168.2.2 icmp_seq=2 timeout
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=61.794 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=47.520 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=62 time=48.322 ms
```

Fig 14: Ping from PC1 to PC2 using RIP routes

Step 23: Leave the debug running for some time. The updates repeat every 30 seconds on both routers (later updates on R2 and R1 shown below).

```
R2#debug ip rip
RIP protocol debugging is on
R2#
*Mar  1 01:13:46.659: RIP: received v1 update from 10.1.1.1 on FastEthernet0/0
*Mar  1 01:13:46.659:      192.168.1.0 in 1 hops
R2#
*Mar  1 01:13:48.635: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.2)
*Mar  1 01:13:48.635: RIP: build update entries
*Mar  1 01:13:48.635:   network 192.168.2.0 metric 1
R2#
*Mar  1 01:13:52.407: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.2.1)
*Mar  1 01:13:52.407: RIP: build update entries
*Mar  1 01:13:52.407:   network 10.0.0.0 metric 1
*Mar  1 01:13:52.407:   network 192.168.1.0 metric 2
R2#
*Mar  1 01:14:14.471: RIP: received v1 update from 10.1.1.1 on FastEthernet0/0
*Mar  1 01:14:14.471:      192.168.1.0 in 1 hops
R2#
*Mar  1 01:14:15.847: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.2)
*Mar  1 01:14:15.847: RIP: build update entries
*Mar  1 01:14:15.847:   network 192.168.2.0 metric 1
R2#
*Mar  1 01:14:20.171: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.2.1)
*Mar  1 01:14:20.171: RIP: build update entries
*Mar  1 01:14:20.171:   network 10.0.0.0 metric 1
*Mar  1 01:14:20.171:   network 192.168.1.0 metric 2
R2#
*Mar  1 01:14:45.299: RIP: received v1 update from 10.1.1.1 on FastEthernet0/0
*Mar  1 01:14:45.299:      192.168.1.0 in 1 hops
*Mar  1 01:14:45.771: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.2)
*Mar  1 01:14:45.771: RIP: build update entries
*Mar  1 01:14:45.771:   network 192.168.2.0 metric 1
R2#
*Mar  1 01:14:48.719: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.2.1)
*Mar  1 01:14:48.719: RIP: build update entries
*Mar  1 01:14:48.719:   network 10.0.0.0 metric 1
*Mar  1 01:14:48.719:   network 192.168.1.0 metric 2
R2#
```

Fig 15: Repeated RIP updates on R2 (continued)

```
*Mar  1 01:15:56.003: RIP: build update entries
*Mar  1 01:15:56.003:   network 10.0.0.0 metric 1
*Mar  1 01:15:56.007:   network 192.168.2.0 metric 2
R1#
*Mar  1 01:16:14.227: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.1)
*Mar  1 01:16:14.227: RIP: build update entries
*Mar  1 01:16:14.227:   network 192.168.1.0 metric 1
R1#
*Mar  1 01:16:15.691: RIP: received v1 update from 10.1.1.2 on FastEthernet0/0
*Mar  1 01:16:15.691:      192.168.2.0 in 1 hops
R1#
*Mar  1 01:16:24.119: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.1.1)
*Mar  1 01:16:24.119: RIP: build update entries
*Mar  1 01:16:24.119:   network 10.0.0.0 metric 1
*Mar  1 01:16:24.123:   network 192.168.2.0 metric 2
R1#
*Mar  1 01:16:44.135: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.1)
*Mar  1 01:16:44.135: RIP: build update entries
*Mar  1 01:16:44.135:   network 192.168.1.0 metric 1
*Mar  1 01:16:44.663: RIP: received v1 update from 10.1.1.2 on FastEthernet0/0
*Mar  1 01:16:44.663:      192.168.2.0 in 1 hops
R1#
*Mar  1 01:16:51.855: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.1.1)
*Mar  1 01:16:51.855: RIP: build update entries
*Mar  1 01:16:51.855:   network 10.0.0.0 metric 1
*Mar  1 01:16:51.859:   network 192.168.2.0 metric 2
R1#
*Mar  1 01:17:10.659: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.1.1.1)
*Mar  1 01:17:10.659: RIP: build update entries
*Mar  1 01:17:10.659:   network 192.168.1.0 metric 1
*Mar  1 01:17:11.299: RIP: received v1 update from 10.1.1.2 on FastEthernet0/0
*Mar  1 01:17:11.299:      192.168.2.0 in 1 hops
R1#
*Mar  1 01:17:21.151: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.1.1)
*Mar  1 01:17:21.151: RIP: build update entries
*Mar  1 01:17:21.151:   network 10.0.0.0 metric 1
*Mar  1 01:17:21.155:   network 192.168.2.0 metric 2
R1#
```

Fig 16: Repeated RIP updates on R1 (continued)

B5. Stop the Debug and Check the RIP Database

Step 24: On R1, stop the debugging and display the RIP database. The route 192.168.2.0/24 is learned [1 hop] via 10.1.1.2 (R2) on Fa0/0; the other entries are directly connected.

```
R1# undebug ip rip
```

```
R1# show ip rip database
```

```
R1#undebug ip rip
RIP protocol debugging is off
R1#show ip rip database
10.0.0.0/8    auto-summary
10.0.0.0/8    directly connected, FastEthernet0/0
192.168.1.0/24    auto-summary
192.168.1.0/24    directly connected, FastEthernet0/1
192.168.2.0/24    auto-summary
192.168.2.0/24
    [1] via 10.1.1.2, 00:00:22, FastEthernet0/0
R1#
```

Fig 17: show ip rip database on R1

Step 25: On R2, display the RIP database. The route 192.168.1.0/24 is learned [1 hop] via 10.1.1.1 (R1) on Fa0/0.

```
R2# show ip rip database
```

```
R2#show ip rip database
10.0.0.0/8    auto-summary
10.0.0.0/8    directly connected, FastEthernet0/0
192.168.1.0/24    auto-summary
192.168.1.0/24
    [1] via 10.1.1.1, 00:00:02, FastEthernet0/0
192.168.2.0/24    auto-summary
192.168.2.0/24    directly connected, FastEthernet0/1
R2#
```

Fig 18: show ip rip database on R2

Step 26: Optional: show ip route on both routers should now list the remote LAN with the code R (RIP), for example R 192.168.2.0/24 [120/1] via 10.1.1.2 on R1. Add your own screenshot here.

```
R1# show ip route
```

Result

✓ EXPERIMENT COMPLETED SUCCESSFULLY

Static routing was first configured on R1 and R2 and then replaced by dynamic routing (RIP) on the same network. In both cases PC1 and PC2 in different networks could communicate through the routers.

VIVA QUESTIONS AND ANSWERS

A. Routing and Static Routing

Q1. What is routing?

Ans: Routing is the process of selecting the path along which a packet travels from the source network to the destination network. A router checks the destination IP of each packet against its routing table and forwards it to the next hop.

Q2. What is static routing?

Ans: In static routing, the network administrator manually enters each route into the router. The route stays the same until it is changed by hand. In this experiment, the ip route command was used on both routers.

Q3. What is dynamic routing? How is it different from static?

Ans: Dynamic routing uses protocols such as RIP, OSPF and EIGRP so routers automatically exchange information and adapt to changes. Static routing is manual, does not adapt if a link fails, but uses no bandwidth or CPU for updates and is simple and predictable.

Q4. What are the advantages of static routing?

Ans: It is simple to configure in small networks, needs no CPU or bandwidth for route exchange, is more secure (no updates can be spoofed) and is predictable because the path is fixed. It also has a low administrative distance of 1, so it is preferred over dynamic routes.

Q5. What are the disadvantages of static routing?

Ans: It does not scale to large networks because every route must be typed on every router. It does not adjust automatically when a link or router fails, so the administrator has to fix it manually, and mistakes are easy to make.

Q6. Where is static routing generally used?

Ans: In small networks, stub networks having only one exit path, for default routes to the internet, as a backup (floating static route) and where security or control is more important than flexibility.

Q7. What is the command for static routing?

Ans: ip route <destination network> <subnet mask> <next-hop IP or exit interface>. In this experiment: ip route 192.168.2.0 255.255.255.0 10.1.1.2 on R1.

Q8. Explain the command ip route 192.168.2.0 255.255.255.0 10.1.1.2.

Ans: It tells R1: “to reach the network 192.168.2.0/24, send the packets to the next-hop router having the address 10.1.1.2”. 192.168.2.0 is the destination network, 255.255.255.0 is its mask and 10.1.1.2 is R2’s interface on the common link.

Q9. What is a next hop?

Ans: The next hop is the IP address of the next router along the path to the destination. For R1, the next hop towards 192.168.2.0 is 10.1.1.2 (R2’s Fa0/0); for R2, the next hop towards 192.168.1.0 is 10.1.1.1 (R1’s Fa0/0).

Q10. Why do we need to configure a route on both routers?

Ans: Ping needs a path both ways. R1 needs a route to 192.168.2.0 for the request to reach PC2, and R2 needs a route to 192.168.1.0 to send the reply back to PC1. If only one side is configured, the ping fails (request goes but reply never returns).

Q11. What if the static route is added only on R1?

Ans: PC1’s packets reach PC2, but PC2’s reply goes to R2, which has no route to 192.168.1.0 and drops it. So the ping gives “Request timed out” or “Destination host unreachable”.

Q12. What is a routing table?

Ans: It is the table inside the router storing the known destination networks, how they were learned, the metric and the next hop/exit interface. It is displayed using the show ip route command.

Q13. What do the codes C and S mean in show ip route?

Ans: C means a directly connected network (10.0.0.0/8 and 192.168.1.0/24 on R1), added automatically when the interface has an IP and is up. S means a static route that was manually configured (192.168.2.0/24 via 10.1.1.2).

Q14. Explain the line "S 192.168.2.0/24 [1/0] via 10.1.1.2".

Ans: S = static route. 192.168.2.0/24 is the destination network. [1/0] shows administrative distance 1 and metric 0. "via 10.1.1.2" is the next hop.

Q15. What is administrative distance (AD)?

Ans: AD is the trustworthiness of the source of a route; a lower value is preferred. Connected = 0, Static = 1, EIGRP = 90, OSPF = 110, RIP = 120. If a router learns the same network from two sources, it uses the one with the lowest AD.

Q16. What is a default route?

Ans: A default route (0.0.0.0/0) is used when no more specific route matches. It is configured as ip route 0.0.0.0 0.0.0.0 <next-hop>. It is often used on stub routers to send all unknown traffic to the ISP.

Q17. What is a floating static route?

Ans: It is a backup static route given a higher administrative distance (e.g. ip route ... 10.1.1.2 150). It stays hidden in the table while the primary route (lower AD) is available and becomes active only if the primary route fails.

Q18. What does "Gateway of last resort is not set" mean?

Ans: It means the router has no default route. Packets for networks not in the routing table are dropped. A default route sets the gateway of last resort.

Q19. What is the difference between next-hop IP and exit interface in a static route?

Ans: With a next-hop IP, the router recursively finds the exit interface for that IP. With an exit interface (ip route ... fa0/0), the router sends the packet out of that interface directly; this is fine on point-to-point links, but on Ethernet it needs ARP for every destination. Using the next hop is preferred on Ethernet.

Q20. What is the longest prefix match?

Ans: When more than one route matches a destination, the router uses the route with the longest subnet mask (most specific). A /24 route will be chosen before a /8 route, and a default route /0 is used only when nothing else matches.

Q21. How can a static route be removed?

Ans: Use the no form of the same command, e.g. no ip route 192.168.2.0 255.255.255.0 10.1.1.2 in global configuration mode.

Q22. What happens to a static route if the exit interface goes down?

Ans: The route is removed from the routing table automatically, because its next hop is no longer reachable. It returns when the interface comes back up.

Q23. How is a static route different from a connected route?

Ans: A connected route appears automatically when an interface is configured and up (AD 0). A static route is manually entered for networks that are not directly attached (AD 1).

B. GNS3 and Devices

Q24. What is GNS3?

Ans: GNS3 (Graphical Network Simulator-3) is software that emulates real Cisco IOS routers and other devices, so a network can be designed and tested without physical hardware. Unlike Packet Tracer, it runs real IOS images.

Q25. What is the difference between GNS3 and Packet Tracer?

Ans: Packet Tracer simulates devices with limited command sets and is easy for beginners. GNS3 emulates actual router images (Dynamips) and supports real IOS commands and features, but needs images and more system resources.

Q26. What is VPCS?

Ans: VPCS (Virtual PC Simulator) is a lightweight PC emulator used in GNS3. It provides basic commands such as ip, ping, trace and show, which is enough for testing connectivity. It needs very few resources.

Q27. What is the format of the ip command in VPCS?

Ans: ip <address> <mask> <gateway>, for example ip 192.168.1.2 255.255.255.0 192.168.1.1. The VPCS then checks for a duplicate address and shows the configuration.

Q28. Why did the VPCS say "Checking for duplicate address..."?

Ans: VPCS sends a gratuitous ARP to find out whether another device already uses the same IP address. This avoids an IP conflict in the network.

Q29. What is the console of a device?

Ans: The console is the terminal window (opened by right-click → Console) where we type commands for the router or VPC. It works like the console port on a real device.

Q30. Why did we start all the devices before configuring?

Ans: The devices in GNS3 are powered off in the beginning. Until they are started (green Start button), there is no console, the routers do not boot and the links remain red. After starting them, the link status points change to green.

Q31. Why do the router interfaces show red dots just after starting?

Ans: The router interfaces are administratively shut down by default ("changed state to administratively down" in the log). After we give an IP and use no shutdown, the interface and link become up (green).

Q32. What is the use of an Ethernet switch in this topology?

Ans: Each switch connects its PC to the router's LAN port. It forwards frames inside the LAN using MAC addresses; no configuration is needed for it in this experiment.

Q33. Why were the routers connected with Fa0/0 to Fa0/0?

Ans: It is the link between the two routers (10.1.1.0/8). Both ends are FastEthernet ports, and a router-to-router link in GNS3 behaves like a cross-over connection.

Q34. Which mode is used to configure the router?

Ans: Global configuration mode, reached by typing conf t from privileged mode (R1#). Interface commands are given after entering a specific interface (interface fa 0/0).

Q35. What does the "no shutdown" command do?

Ans: It enables the interface administratively. Router interfaces are shut down by default; without no shutdown the port stays down even with correct cabling and IP.

Q36. What does show ip interface brief show?

Ans: A short table of every interface, its IP address, whether it is configured manually, its status (up/down) and the line protocol. In this experiment, both interfaces of R1 and R2 are up/up, which confirms that the configuration is right.

Q37. What is the difference between "Status" and "Protocol" in show ip interface brief?

Ans: Status is the physical (Layer-1) state: is the cable and port active. Protocol is the Layer-2 state: is the line protocol (e.g. Ethernet keepalive) working. Both must be up for the interface to carry traffic.

Q38. What does "administratively down" mean?

Ans: The interface was turned off by the shutdown command (or the default state). It will remain down until no shutdown is issued.

Q39. What is the purpose of the “end” command?

Ans: It exits any configuration mode and goes straight back to privileged EXEC mode (R1#). It is the same as pressing Ctrl+Z.

C. Errors, Troubleshooting and Verification

Q40. Why did we get “Bad mask /8 for address 10.0.0.0”?

Ans: 10.0.0.0/8 is the network address of the class A network (all host bits zero), so it can not be assigned to an interface. We used the valid host address 10.1.1.1 with the mask 255.0.0.0 instead.

Q41. What is the “Duplicate address 10.1.1.1 on FastEthernet0/0” message?

Ans: The router detected that another device (here R1, source MAC c001.2704.0000) already uses 10.1.1.1. Two devices cannot have the same IP in a network; the problem was solved by giving R2 the address 10.1.1.2.

Q42. Why did “Translating "end"...” and “Invalid input detected” appear?

Ans: “end” was typed in the privileged mode where it is not a command, so the router tried to treat it as a hostname and look it up in DNS. The “Invalid input” appeared for “clear”, which is not valid in interface mode. These were harmless typing errors.

Q43. What is the “%SYS-5-CONFIG_I” message?

Ans: It is an information message saying that the configuration was changed from the console. It is displayed after every exit from configuration mode.

Q44. Why must both ends of the router link be in the same network?

Ans: The two interfaces must be in the same subnet to be able to talk directly. Here 10.1.1.1 and 10.1.1.2 are both in 10.0.0.0/8. Otherwise the next hop would be unreachable and the static route would not work.

Q45. What if the next-hop IP in the static route is wrong?

Ans: The router cannot reach the next hop, so the route is not installed or packets are lost. The ping from the PC would fail. The route must point to the neighbouring router’s IP on the common link.

Q46. What if the mask in the static route is wrong?

Ans: The router would use a different network range for the route. The destination might not be matched, or too many addresses could be matched, so ping may fail or other traffic may go the wrong way.

Q47. What if the PC’s gateway is wrong?

Ans: The PC will not be able to send packets outside its network, because it sends them to a wrong or non-existent address. Even if the routers are perfect, ping to the other network fails.

Q48. What if an interface is left in shutdown?

Ans: The connected network does not appear in the routing table and the static route that uses it as a next hop cannot work. The ping fails until no shutdown is done.

Q49. How do you verify a static route?

Ans: Use show ip route to see the S entry, and ping/trace from the PC to the remote network. A successful ping from PC1 to PC2 and back proves both routers are correctly configured.

Q50. What is the purpose of the trace command?

Ans: trace (tracert/traceroute) shows each router along the path to the destination by sending packets with increasing TTL values. In our topology, it would show R1 (192.168.1.1), R2 (10.1.1.2) and then PC2.

Q51. Why does the first ping often time out?

Ans: The first packet needs ARP resolution on each LAN to find the MAC address of the gateway/destination, which takes longer than the ping timeout. The following pings succeed as the ARP entries are cached.

Q52. What does TTL do in the ping reply?

Ans: TTL (Time To Live) is decreased by every router the packet passes and prevents packets from looping forever. When PC1 pings PC2, the TTL in the reply is reduced by two (two routers crossed).

Q53. How would you save the router configuration?

Ans: Use write memory (or copy running-config startup-config) in privileged mode. Otherwise the running configuration in RAM is lost when the router is restarted.

Q54. What if there were 3 routers in a line? How many static routes?

Ans: Each router needs a route to every network not directly connected. The end routers need routes to the remote LANs through their only neighbour, and the middle router needs routes to both end LANs. That is why static routing becomes difficult as the network grows.

Q55. Can we use a static route and RIP together? Which one is used?

Ans: Yes. If both give a route to the same network, the router prefers the static route because its administrative distance (1) is lower than RIP's (120).

Q56. Which layer does a router work on?

Ans: Layer 3 (Network layer) of the OSI model, as it makes forwarding decisions using IP addresses.

D. Dynamic Routing and RIP

Q57. What is dynamic routing?

Ans: In dynamic routing, routers run a routing protocol and automatically exchange information about the networks they know. Each router builds and updates its own routing table, so no one has to type the routes by hand and the tables change when the network changes. In Part B, RIP was used for this.

Q58. Why did we move from static to dynamic routing?

Ans: Static routes have to be typed on every router and do not adapt if a link fails. With dynamic routing the routers learn the remote LANs themselves, which is easier in larger networks and reacts to changes. It also showed the difference between the two methods on the same topology.

Q59. What are the types of dynamic routing protocols?

Ans: Distance-vector protocols (RIP, IGRP, EIGRP), which tell neighbours their routing table; link-state protocols (OSPF, IS-IS), where every router builds a map of the network; and path-vector (BGP) used between autonomous systems. RIP is a distance-vector protocol.

Q60. What is RIP?

Ans: Routing Information Protocol is a simple distance-vector protocol for small networks. Every router sends its entire routing table to its neighbours at regular intervals, and the best route is the one with the least hops.

Q61. Which metric does RIP use? What is the maximum?

Ans: RIP uses hop count, the number of routers on the path. The maximum is 15; a metric of 16 means unreachable. In our output, 192.168.2.0 is "1 hop" from R1 because R2 is one router away.

Q62. How often does RIP send updates?

Ans: Every 30 seconds. In the debug output, R2 sent its updates through Fa0/0 at 01:13:48, 01:14:15 and 01:14:45, which is about 30 seconds apart (a little random jitter is added to avoid synchronisation).

Q63. What is the administrative distance of RIP and of static routes?

Ans: RIP is 120 and a static route is 1. A lower value is more trusted, so if both a static and a RIP route to the same network exist, the static one wins. That is why the static routes were removed before configuring RIP.

Q64. Why did we remove the static routes before RIP?

Ans: A static route (AD 1) beats a RIP route (AD 120) for the same network. If we left it, the router would keep using the static route and we could not be sure that RIP works. After removal, the routing table showed only connected (C) routes.

Q65. What does the command “router rip” do?

Ans: It starts the RIP routing process on the router and enters the router configuration mode (config-router). The network commands given after it decide which interfaces take part in RIP.

Q66. What does the network command do in RIP?

Ans: It names a directly connected network: all the interfaces which have an IP in that network start sending and receiving RIP updates, and that network is advertised to neighbours. We typed network 10.1.1.0 and network 192.168.1.0 on R1.

Q67. Do we advertise remote networks in the network command?

Ans: No. Only the directly connected networks of that router are entered. The remote networks are learned from the neighbour’s updates. R1 entered 10.1.1.0 and 192.168.1.0, not 192.168.2.0.

Q68. Why was 10.1.1.0 shown as 10.0.0.0 in the outputs?

Ans: RIPv1 is classful. 10.1.1.0 belongs to Class A, so IOS converts the entered network to its classful form 10.0.0.0 (/8). That is why the debug shows “network 10.0.0.0” and the database shows 10.0.0.0/8.

Q69. What is the purpose of “no router rip” before “router rip”?

Ans: It deletes the existing RIP process and all its network statements, giving a clean start. It is optional but removes any old or wrong configuration.

Q70. Which RIP version was used? How do you know?

Ans: Version 1. No version command was given, and the debug lines say “sending v1 update” and “received v1 update”. Cisco routers by default send version 1 and receive both versions.

Q71. What is the difference between RIPv1 and RIPv2?

Ans: RIPv1 is classful (no subnet mask in updates), uses broadcast (255.255.255.255) and has no authentication. RIPv2 is classless (supports VLSM), uses multicast 224.0.0.9 and supports authentication. To use v2 we type version 2 in router rip mode.

Q72. How do we change RIP to version 2?

Ans: In router configuration mode type version 2 (and usually no auto-summary). Both routers should then send and receive v2 updates to multicast 224.0.0.9.

Q73. Why are RIPv1 updates sent to 255.255.255.255?

Ans: RIPv1 uses a broadcast to reach all neighbours on the link, which is seen in the debug output (“sending v1 update to 255.255.255.255”). All devices on the segment receive it, but only RIP routers process it.

Q74. What is auto-summary?

Ans: The router summarises subnets into their classful network when advertising across a different major network. In the show ip rip database output, “auto-summary” entries appear for 10.0.0.0/8, 192.168.1.0/24 and 192.168.2.0/24. It can be switched off with no auto-summary in RIPv2.

Q75. What type of algorithm does RIP use?

Ans: Distance-vector, based on the Bellman-Ford algorithm. A router knows only the distance (hops) and direction (next hop) to a network, as told by neighbours, and not the whole topology.

Q76. What are the advantages of dynamic routing?

Ans: Automatic route discovery, less administrative work, automatic adaptation to link failures and easy scalability. Disadvantages are CPU and bandwidth use for updates, and the need for care about loops, convergence and security.

Q77. Why is RIP not suitable for large networks?

Ans: Its limit is 15 hops, it sends the whole table every 30 seconds (wasting bandwidth), converges slowly and uses only hop count (ignores speed). Larger networks use OSPF or EIGRP.

Q78. What is convergence?

Ans: The time and state in which all routers have the same, correct routing information after the start or a change. After enabling RIP on both routers, the first updates (about 30 seconds) were needed before PC1 could ping PC2.

Q79. What are split horizon and route poisoning?

Ans: Split horizon means a router does not advertise a route back out of the interface from which it learned it. Route poisoning advertises a failed network with metric 16. Both prevent routing loops and count-to-infinity.

Q80. Where can split horizon be seen in our debug output?

Ans: R2 learned 192.168.1.0 from R1 through Fa0/0, so when R2 sends its update out of Fa0/0, it only includes 192.168.2.0, and not 192.168.1.0. Updates through the LAN side (Fa0/1) include 10.0.0.0 and 192.168.1.0.

Q81. What is the count-to-infinity problem?

Ans: When a network fails, routers may keep passing the route to each other and increase the metric step by step until it reaches 16 (infinity for RIP). Split horizon, hold-down timers and route poisoning limit this.

Q82. What are the RIP timers?

Ans: Update 30 s, invalid 180 s (route marked unreachable if no update), hold-down 180 s and flush 240 s (route removed). They can be seen using show ip protocols.

E. Reading the Dynamic Routing Outputs

Q83. What does the command "debug ip rip" do?

Ans: It displays RIP messages in real time as they are sent and received, with the networks and metrics inside. It is useful for learning and troubleshooting, but it can overload a router, so it must be switched off afterwards.

Q84. How do we stop the debugging?

Ans: Use undebug ip rip (or undebug all, or no debug all). The router prints "RIP protocol debugging is off", as seen in the output on R1.

Q85. Explain the line "RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (192.168.1.1)".

Ans: R1 is broadcasting a version-1 RIP update from its interface Fa0/1 whose IP is 192.168.1.1. The following lines list the networks put in this update, with the metric as seen from the receiving side.

Q86. Why does R1 advertise "192.168.2.0 metric 2" on Fa0/1?

Ans: R1 learned 192.168.2.0 from R2 with 1 hop, and adds 1 for the extra hop before telling its LAN side. So the metric shown in the update is 2. The directly connected 10.0.0.0 is advertised with metric 1.

Q87. Explain "RIP: received v1 update from 10.1.1.1 on FastEthernet0/0 – 192.168.1.0 in 1 hops".

Ans: R2 got a RIP update from its neighbour R1 (10.1.1.1) on Fa0/0. The update says that network 192.168.1.0 is 1 hop away through R1, so R2 can install a route to it with next hop 10.1.1.1.

Q88. What do the "network ... metric 1/2" lines mean?

Ans: They are the routes included in the RIP update and the hop count assigned to each. Directly connected networks have metric 1 in the update, and a route learned from a neighbour is one more.

Q89. Why were the first pings timed out in the screenshot?

Ans: The routers needed to exchange their first RIP updates (up to 30 seconds) and the hosts needed ARP resolution. After the routes were learned, the replies arrived. The first two requests timed out and the next three succeeded.

Q90. Why is the TTL 62 in the ping reply?

Ans: VPCS starts packets with TTL 64, and each router lowers it by 1. The reply passed through two routers (R2 and R1), so 64 − 2 = 62. It proves that the traffic was routed through both routers.

Q91. Why is the ping time (about 48 ms) higher than a normal LAN?

Ans: GNS3 emulates real router software on a computer, so the packets are processed by emulated devices and virtual links. Delays of tens of milliseconds are normal there.

Q92. What does the command show ip rip database show?

Ans: It lists the RIP routing database: the networks known, whether they are directly connected or learned via a neighbour, the metric in brackets, the neighbour IP, the age of the update and the exit interface.

Q93. Explain "192.168.2.0/24 [1] via 10.1.1.2, 00:00:22, FastEthernet0/0".

Ans: On R1, the network 192.168.2.0/24 was learned from RIP with metric 1 hop; the next hop is 10.1.1.2 (R2); the last update was 22 seconds ago; and the packets leave through Fa0/0.

Q94. What does "directly connected, FastEthernet0/0" mean in the RIP database?

Ans: The network 10.0.0.0/8 is attached to the router's own interface, so it is a connected network which RIP is advertising, and not a route learned from a neighbour.

Q95. How is the routing table different after RIP is configured?

Ans: In addition to the C (connected) routes, a line with the code R appears for the remote LAN, e.g. R 192.168.2.0/24 [120/1] via 10.1.1.2. The 120 is RIP's administrative distance and the 1 is the hop count.

Q96. Which commands are used to verify RIP?

Ans: show ip route (R routes), show ip protocols (timers, networks, neighbours), show ip rip database, debug ip rip and finally ping/trace between the PCs.

F. Static vs Dynamic and Scenarios

Q97. Compare static and dynamic routing.

Ans: Static: manual, no overhead, no automatic adaptation, simple, AD 1, best for small or stub networks. Dynamic: automatic, uses CPU/bandwidth, adapts to changes, scalable, AD depends on protocol (RIP 120), best for medium and large networks.

Q98. What if we configure both a static route and RIP for the same network?

Ans: The router installs the static route because of its lower administrative distance (1 versus 120). The RIP route is kept in the RIP database but is not used until the static route is removed.

Q99. What if a network statement is missing on R2?

Ans: R2 will not advertise that network (and will not send/receive updates on the interface). R1 would not learn it, so the pings to it from the other LAN fail.

Q100. What if the network statement is wrong (e.g. network 192.168.3.0)?

Ans: No interface is in that network, so RIP does not run on the intended interface. The network is not advertised and the neighbour does not learn it. The correct directly connected network must be entered.

Q101. What if one router runs RIP and the other does not?

Ans: The routers do not exchange routes. The router with RIP sends updates but nobody listens, and the other router learns nothing, so neither LAN can reach the other.

Q102. What if the link between R1 and R2 is cut?

Ans: The routers stop receiving updates. After 180 seconds the learned route becomes invalid and after 240 seconds it is flushed, so ping between the LANs fails. With static routes it would remain in the table (until the interface goes down).

Q103. What if the two ends of the link are not in the same network?

Ans: The routers cannot be neighbours and do not exchange RIP updates. Both ends must be in the same subnet, here 10.1.1.1 and 10.1.1.2 in 10.0.0.0/8.

Q104. Why is it important to turn off the debug?

Ans: Debug generates a message for every event and uses CPU. On a busy router it can slow it or even make it unreachable. So we switch it off with undebug ip rip after observing.

Q105. Can RIP load balance?

Ans: Yes, RIP can use up to 4 equal-cost paths by default (maximum path: 4), splitting the traffic between them. It is shown in show ip protocols.

Q106. What is a passive interface in RIP?

Ans: An interface where RIP updates are not sent but received. It is used on the LAN side to avoid sending updates to hosts that do not need them, using passive-interface fa0/1 in router rip mode.

Q107. How could this setup be improved for a real network?

Ans: Use RIPv2 with no auto-summary, set passive-interface on LAN ports, add authentication, or use OSPF/EIGRP for faster convergence and bigger networks.

Q108. Which is better for this small experiment, static or dynamic?

Ans: For two routers with one path, static is simple and enough. Dynamic with RIP has the advantage that it adjusts automatically and does not need manual changes when networks are added. The experiment is done to show both approaches.